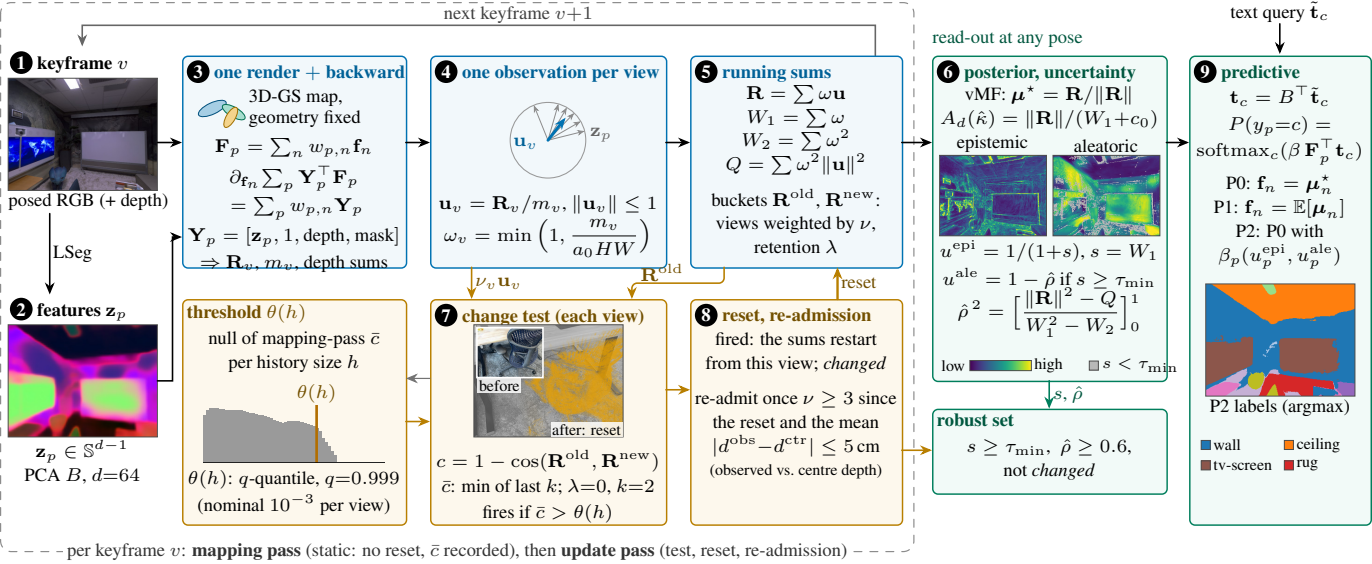

next keyframe $v{+}1$
text query $\tilde{\mathbf{t}}_c$
read-out at any pose
1 keyframe v
posed RGB (+ depth)
LSeg
2 features $\mathbf{z}_p$
$\mathbf{z}_p \in \mathbb{S}^{d-1}$
PCA B, d=64
3 one render + backward
3D-GS map, geometry fixed
$\mathbf{F}_p = \sum_n w_{p,n}\mathbf{f}_n$
$\partial_{\mathbf{f}_n}\sum_p \mathbf{Y}_p{}^\top\mathbf{F}_p$
$= \sum_p w_{p,n}\mathbf{Y}_p$
$\mathbf{Y}_p = [\mathbf{z}_p, 1, \text{depth}, \text{mask}]$
$\Rightarrow \mathbf{R}_v, m_v$, depth sums
4 one observation per view
$\mathbf{u}_v = \mathbf{R}_v/m_v$, $\|\mathbf{u}_v\| \le 1$
$\omega_v = \min\left(1, \frac{m_v}{a_0 HW}\right)$
5 running sums
$\mathbf{R} = \sum \omega\mathbf{u}$
$W_1 = \sum \omega$
$W_2 = \sum \omega^2$
$Q = \sum \omega^2\|\mathbf{u}\|^2$
buckets $\mathbf{R}^{\text{old}}$, $\mathbf{R}^{\text{new}}$: views weighted by ν, retention λ
6 posterior, uncertainty
vMF: $\boldsymbol{\mu}^\star = \mathbf{R}/\|\mathbf{R}\|$
$A_d(\hat\kappa) = \|\mathbf{R}\|/(W_1{+}c_0)$
epistemic
aleatoric
$u^{\text{epi}} = 1/(1{+}s)$, $s = W_1$
$u^{\text{ale}} = 1 - \hat\rho$ if $s \ge \tau_{\min}$
$\hat\rho^{\,2} = \left[\frac{\|\mathbf{R}\|^2 - Q}{W_1^2 - W_2}\right]_0^1$
low
high
$s < \tau_{\min}$
9 predictive
$\mathbf{t}_c = B^\top\tilde{\mathbf{t}}_c$
$P(y_p{=}c) = \text{softmax}_c(\beta\,\mathbf{F}_p^\top\mathbf{t}_c)$
P0: $\mathbf{f}_n = \boldsymbol{\mu}_n^\star$
P1: $\mathbf{f}_n = \mathbb{E}[\boldsymbol{\mu}_n]$
P2: P0 with $\beta_p(u_p^{\text{epi}}, u_p^{\text{ale}})$
P2 labels (argmax)
wall
ceiling
tv-screen
rug
$\nu_v\mathbf{u}_v$
$\mathbf{R}^{\text{old}}$
reset
threshold $\theta(h)$
null of mapping-pass $\bar c$ per history size h
$\theta(h)$
$\theta(h)$: q-quantile, q=0.999
(nominal 10^{-3} per view)
7 change test (each view)
before
after: reset
$c = 1 - \cos(\mathbf{R}^{\text{old}}, \mathbf{R}^{\text{new}})$
$\bar c$: min of last k; λ=0, k=2
fires if $\bar c > \theta(h)$
8 reset, re-admission
fired: the sums restart from this view; *changed*
re-admit once $\nu \ge 3$ since the reset and the mean
$|d^{\text{obs}} - d^{\text{ctr}}| \le 5$ cm
(observed vs. centre depth)
$s, \hat\rho$
robust set
$s \ge \tau_{\min}$, $\hat\rho \ge 0.6$, not *changed*
per keyframe v: **mapping pass** (static: no reset, $\bar c$ recorded), then **update pass** (test, reset, re-admission)